

Homework 6

Solutions must be **typed** and **submitted electronically**.

1. Show that a 1-round keyed Feistel cipher **cannot** be a secure PRP, no matter what its round functions are. That is, construct a distinguisher that successfully distinguishes $\mathcal{L}_{\text{prp-real}}^F$ and $\mathcal{L}_{\text{prp-rand}}^F$, knowing only that F is a 1-round Feistel cipher. In particular, the purpose is to attack the Feistel transform and not its round function, so your attack should work no matter what the round function is.
2. Show that a 2-round keyed Feistel cipher **cannot** be a secure PRP, no matter what its round functions are. Your attack should work without knowing the round keys, and it should work even with different (independent) round keys.

Hint: A successful attack requires two queries.

3. Let F be a secure PRF with λ -bit outputs, and let G be a PRG with stretch ℓ . Define

$$F'(k, r) = G(F(k, r)).$$

So F' has outputs of length $\lambda + \ell$. Prove that F' is a secure PRF.